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FEDERAL SPECIFICATION SHEET

PLUMBING FIXTURES (SHOWER BATH AND EMERGENCY EYE AND FACE WASH OUTFITS) (DETAIL SPECIFICATION)

This specification forms a part of the latest issue of Federal Specification WW-P-541/GEN.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers specific requirements for shower bath, emergency eye, and emergency eye and face wash outfits and components.

1.2 Classification.

1.2.1 Parts, types, and classes. Shower bath outfits and components of outfits shall be of the following parts, types, and classes as specified (see 6.1).

Part A - Shower installation outfits.

Type I - Shower cabinets, free standing.

Class 1 - Single unit.
Class 2 - Single unit complete with receptor.
Class 3 - Multiple unit.
Class 4 - Combination shower and dressing unit.

Type II - Emergency showers.
Type III - Wall-mounted showers.
Type IV - Column type showers.
Type V - Bathtub showers.
Type VI - Special.

Part B - Shower receptors.

Part C - Shower heads.

Type I - Heads for shower bathing.

Class 1 - Nonadjustable spray.
Class 2 - Adjustable spray.

Type II - Heads for emergency showers.
Type III - Special.

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Part D - Emergency eye, and emergency eye and face wash.

Type I - Emergency eye wash.
Type II - Emergency eye and face wash.
Type III - Special.

Part E - Valves.

Type T - Thermostatic mixing.
Type P - Pressure balancing.
Type M - Mechanical.

Class 1 - Mixing valve.
Class 2 - Single control valve.

Type C - Combined supply, separate, hot and cold water valves.

Part F - Drains.

1.2.2 Definitive part number (DPN). DPNs have not been established for the plumbing fixtures covered by this specification sheet. The items covered herein are not normally stocked and issued, but rather are ordered for specific application and require that option decisions be made regarding specific installation criteria. This is best controlled through use of ordering data options (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Latest issue of WW-P-541/GEN and documents referenced therein.

3. REQUIREMENTS

3.1 Materials and components. Materials and components shall conform to the basic Specification WW-P-541/GEN, and as hereinafter specified.

3.2 Part A, shower installations (outfits). Shower outfits furnished under this specification shall include the fixture and the associated fittings and trim necessary to make a complete installation from wall or floor connections to rough piping.

3.2.1 Materials.

3.2.1.1 Single units. Shower cabinets shall be porcelain enameled steel, baked enamel coated steel, stainless steel, or plastic. Material for plastic showers shall conform to ANSI Z124.2. Curtain rod (tube) shall be stainless steel, copper alloy, or aluminum alloy. Aluminum alloy shall be 3000 or 6000 series, and temper shall be at the manufacturer's option.

3.2.1.2 Multiple unit, and combination shower and dressing unit. Partitions for multistall showers and combination shower and dressing room shall be porcelain enameled steel, baked enamel coated steel, stainless steel or plastic. Pilasters and posts shall be of the same material as panels. Top rails joining pilasters shall be stainless steel or aluminum alloy of any suitable composition and temper.

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3.2.1.3 Emergency showers. Partitions shall be steel sheet. Coating on partitions shall be at the manufacturer's option. Pipe and floor grate shall be zinc coated or shall have any other protective coat at the manufacturer's option. Type of steel for pipe and grate shall be at the manufacturer's option.

3.2.1.4 Wall mounted showers. Showers shall be stainless steel, or baked enamel coated on carbon steel.

3.2.1.5 Column mounted showers. The column shall be stainless steel.

3.2.1.6 Bathtub and other showers.

3.2.1.6.1 Tub spouts. Spouts shall be copper alloy, die-cast zinc alloy, or molded plastic.

3.2.1.6.2 Shower doors. Frames for shower doors shall be any of the aluminum magnesium-silicon alloys, series 6000, and temper at the manufacturer's option. Plastic panels shall be an acrylic plastic sheet or any other suitable plastic. Glass panels shall be fully tempered glass.

3.2.1.6.3 Escutcheons. Escutcheons shall be stainless steel, copper alloy, die-cast zinc alloy, or molded plastic.

3.2.1.6.4 Pipe and tube. Pipe shall be copper alloy, and tube between shower connections and control valves shall be copper or copper alloy.

3.2.2 Type I, classes 1 and 2 shower cabinets.

3.2.2.1 Size. The size of steel cabinets and plastic cabinets complete with receptor, shall be as specified (see 6.1 and 6.2).

3.2.2.2 Base metal. Thickness of cabinet walls shall be in accordance with the manufacturer's standard commercial product.

3.2.2.3 Parts. Thickness of parts including corner members and top frames shall be 0.043 inch or greater (see 4.4.1.1).

3.2.2.4 Closure strips, and receptor flange. Thickness of closure strips shall be 0.032 inch or greater, and thickness of receptor flange shall be 0.043 inch or greater (see 4.4.1.1).

3.2.2.5 Coating on steel cabinets. The total thickness of porcelain enamel shall be 5 mils (0.005 inch) or greater for porcelain enameled steel showers. For baked enamel steel units the thickness of the base coat and final coat shall be as follows (see 4.4.1.1):

(a) Base coat. Thickness of zinc base or equal coat shall be 1 mil (0.001 inch), or 0.625 ounce per square foot on each side of sheet steel. Zinc or iron phosphate base coat shall be used, and the thickness of the phosphate primer shall be 100 mils per square foot or greater for zinc phosphate and 20 mils or greater for iron phosphate on each side of the sheet steel. Thickness of the primers shall be 0.1 mil (0.0001 inch) or greater.

(b) Enamel. Thickness of the baked enamel shall be 1.5 mils (0.0015 inches) minimum.

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3.2.2.6 Plastic shower unit. Plastic units shall meet the performance requirements of ANSI Z124.2. Units may be integrally molded in one piece or made in sections for field assembly.

3.2.2.7 Curtain rod or door. Shower cabinets shall have a curtain rod or door, as specified (see 6.1).

3.2.2.7.1 Curtain rod. Outside diameter of the tube shall be 3/4 inch or greater and wall thickness shall be 0.025 inch or greater (see 4.4.1.2). The tube shall be located behind pilasters and threshold, and below the top of the cabinet.

3.2.2.7.2 Door. Design of the shower door shall be at the manufacturer's option. Door shall have a plastic or glass panel. Panels shall be translucent and pattern of glass and plastic panels shall be at the manufacturer's option. Color of plastic panels shall be at the manufacturer's option.

3.2.2.7.3 Entry. Door or curtain rod shall be located for front entry, but when specified (see 6.1), shall be located for corner entry.

3.2.2.8 Shower cabinet fittings and trim. Cabinets shall have concealed or exposed fittings and trim, as specified (see 6.1).

3.2.2.9 Soap dish. Steel shower cabinets shall have a chromium plated steel, enameled steel, stainless steel, or plastic soap dish at the manufacturer's option. Soap dish may be integrally molded in plastic showers.

3.2.2.10 Finish: cabinets, fittings, and trim, classes 1 and 2.

3.2.2.10.1 Cabinets. Walls and sides shall be porcelain enameled, coated with a baked enamel finish or stainless steel. Color of the porcelain enamel or the baked enamel shall be white or gray. The value of the color shall be at the manufacturer's option. Stainless steel shall have a No. 4 general-purpose polished finish.

3.2.2.10.2 Soap dish, curtain rod, and shower door. Chromium plated soap dishes, curtain tubes, door frame, and door hardware shall have a bright finish. Stainless steel soap dishes, door frames, and door hardware shall have a No. 4 general-purpose polished finish. Plastic soap dishes may have a molded finish.

3.2.3 Type I, classes 3 and 4, multiple, and combination showers.

3.2.3.1 Partitions. Partitions shall be porcelain enameled, coated with baked enamel finish or stainless steel. Color of the porcelain enamel or the baked enamel shall be white or gray. The value of the color shall be at the manufacturer's option. Stainless steel shall have a No. 4 general-purpose polished finish.

3.2.3.2 Thickness of panels, pilasters, posts, rails, and columns.

3.2.3.2.1 Panels. Thickness of panels shall be 1 $\pm$1/16 inch. Construction shall consist of two metal panels 0.036 inch or greater in thickness with a core material sandwiched between the panels (see 4.4.1.3). Type of core material and shape of cores shall be at the manufacturer's option.

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3.2.3.2.2 Pilasters, posts, and rails. Thickness of uncoated pilasters, posts and top rails shall be 0.044 inch or greater (see 4.4.1.3).

3.2.3.2.3 Columns. Column thickness shall be as specified in 3.2.6.1.3.

3.2.3.3 Pilaster and post construction. Pilasters and posts shall be formed from steel. Joints between top rails and pilasters shall be welded or joined with suitable stainless steel fasteners.

3.2.3.4 Partition supports and hardware. Showers shall be supported by a framework including a curtain rail attached to radial braces and vertical outside standards. Tubes of circular, square or any other suitable cross-section shall be used for framework, rails, braces, and standard. Arc of curtain rail shall be continuous for four or more person showers. For three or less persons, shower wall brackets shall be included for securing curtain rail to wall. Vertical outside standards shall extend through the floor, or held by a suitable floor bracket. Stainless steel escutcheons shall be provided for each standard. Joints between curtain rail, radial braces and standards shall be welded or joined with suitable stainless steel fasteners.

3.2.3.5 Class 3, multiple showers. Arrangement of partitions, number and location of shower units shall be as specified (see 6.1).

3.2.3.6 Class 4, combination shower and dressing room. Arrangement of partitions, number of shower and dressing units, and location of shower and dressing units, shall be as specified (see 6.1). Dressing unit shall include a bright chromium-plated robe hook and a maple, or any other species of hard wood, or particleboard seat sized to 1 ±1/8 inch thick, width shall be 9-5/8 ±1.8 inches and the length shall be not less than 30 inches nor greater than 35 inches. Top outer edges shall be rounded. The seat shall be fastened to panel using metal brackets. Additional supports for seats such as metal legs shall be included when specified (see 6.1). Finish of seat shall be at the manufacturer's option. Brackets shall be zinc coated or painted with a baked enamel coat.

3.2.4 Type II, emergency shower cabinets. Thickness of steel sheet for panels shall be 0.027 inch or greater (see 4.4.1.3). Construction details for emergency showers shall be as specified (see 6.1).

3.2.5 Type III wall-mounted showers.

3.2.5.1 Single shower units. Single units shall be designed for mounting on walls or in corner of walls, as specified (see 6.1).

3.2.5.2 Semi-circular and three panel showers. Units with two showers shall have the shower and control valves located adjacent to the center to permit installation of a dividing shower partition. Each shower shall be equipped with a soap dish or shelf, or liquid soap dispensers, as specified (see 6.1). Thickness of panels for semi-circular and three panel showers shall be 0.045 inch or greater. Stainless steel panels shall have a No. 4 general-purpose polished finish. Galvanized steel panels shall be coated with a baked enamel finish. Thickness of enamel shall be 1 mil (0.001 inch) or greater (see 4.4.1.3).

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3.2.5.3 In-line shower panel, wall mounted. Showers shall have one to six shower stations, as specified (see 6.1). Units with two or more showers shall have showers and valves located on 36 inch centers. Each shower shall be equipped with a soap dish or liquid soap dispenser, as specified (see 6.1). Thickness of panels shall be 0.070 inch or greater (see 4.4.1.3). Thickness of access panels shall be 0.035 inch or greater (see 4.4.1.3). Stainless steel panels shall have a No. 4 general-purpose polish finish with polish marks in a horizontal direction. Units shall be an open back box form, 4 inches deep and 14 inches high. All exposed edges of metal shall be smooth, deburred, and polished.

3.2.5.4 In-line shower panels, back to back. Showers shall have 2, 4, 6, 8, 10, or 12 shower stations, as specified (see 6.1). Showers shall be mounted wall both ends, wall to stanchion, stanchion to stanchion, or installed on top of dwarf wall, as specified (see 6.1). Units with four or more showers shall have showers and valves located on 36 inch centers. Each shower shall be equipped with soap dish or liquid soap dispenser as specified (see 6.1). Thickness of panels and stanchions shall be 0.070 inch or greater (see 4.4.1.3). Thickness of access panels shall be 0.035 inch or greater (see 4.4.1.3). Stainless steel panels and stanchions shall have a No. 4 general purpose finish with polish marks in a horizontal direction. Stanchion shall have polish marks in a vertical direction. Bottom of stanchion shall have a base flange 0.130 inch or greater thickness with internal bracing of 0.120 inch or greater thickness, extending 3-1/2 inches on three sides into stanchion. Units shall be a closed box from 4-1/2 inches deep and 14 inches high. All exposed edges of metal shall be smooth, deburred, and polished.

3.2.5.5 Control valves. Control valves shall be mounted within the panel for each shower head, or mounted in a single separate cabinet for remote control, as specified (see 6.1). Valves shall be of the following types, as specified (see 6.1):

(a) Separate valves for hot and cold water.
(b) Pressure balancing mixing valve with single handle.
(c) Self-closing valves.
(d) Push button metering valve.
(e) Thermostatic mixing valve.
(f) Single control valves.

Type of handle on valves shall be at the option of the manufacturer.

3.2.6 Type IV, column type construction showers. Showers shall be as follows and as specified (see 6.1):

(a) Shower completely around the column for 4, 5, and 6 shower heads and controls, as specified (see 6.1).
(b) Shower column for location adjacent to a wall with 3 shower heads and controls.
(c) Shower column for location in corner of wall with 2 shower heads and controls.
(d) Multistall showers with radial partitions for 2, 3, 4, 5, and 6 showers.
(e) Combination shower and dressing room with 2 or 4 shower enclosures each and with number of dressing rooms, as specified (see 6.1).

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3.2.6.1 Columns. Columns shall be seam welded.

3.2.6.1.1 Height. Height of columns measured from floor level to shower heads shall be 5, 5-1/2, or 6 feet, as specified (see 6.1).

3.2.6.1.2 Diameter. Outside diameter of column shall be 6 inches, minimum.

3.2.6.1.3 Thickness. Thickness of stainless steel columns, and carbon steel columns with galvanized coating shall be 0.063 inch or greater (see 4.4.1.3).

3.2.6.1.4 Finish, column showers. Stainless steel columns shall have a No. 4 general purpose polished finish.

3.2.6.1.5 Top of columns. The top of columns shall have removable cover plate, or extension tube to ceiling, as specified (see 6.1). When extension tubes are specified (see 6.1), include ceiling height dimension. Top of columns without tube extension shall have a removable cover plate. The truncated section shall be suitable for accommodating the shower heads.

3.2.6.1.6 Bottom of columns. Bottom of columns shall have a flange of not less than 13 gage (0.093 inch) thick attached to a cast iron drain base. Drain assembly configurations shall be proportional to maximum shower head discharge. When shower head flow exceeds 25 gallons per minute (gpm), high-capacity dome-gate drain with capacity up to 60 gpm shall be specified (see 6.1). A removable cover plate for access to bottom supply, drain and vent connection shall be included near the base of the column.

3.2.6.2 Control valves. Control valves shall be mounted into a manifold contained within the column for each shower head, or mounted in a single separate cabinet for remote control, as specified (see 6.1). Valves shall be of the following types as specified (see 6.1):

(a) Separate valves for hot and cold water.
(b) Pressure balancing mixing valves with single handle.
(c) Self-closing valves.
(d) Push button metering valves.
(e) Thermostatic mixing valve for remote control.
(f) Single control valves.

Type of handles on valves shall be at the manufacturer's option.

3.2.7 Type V, bathtub showers and tub spout.

3.2.7.1 Shower doors and curtain rail. Bathtub showers shall have curtain rail or sliding doors, as specified (see 6.1).

3.2.7.1.1 Shower curtain rod. Curtain shall be supported by a tube 1 inch outside diameter or greater with a wall thickness of 0.025 inch or greater (see 4.4.1.2). Tube shall be straight or angled depending upon the shape of the bathtub. Two flanges shall be included with suitable fasteners for attachment to walls and for preventing rotation of the tube. Copper alloy rods and brackets shall have a bright chromium plated finish, and stainless steel rods and brackets shall have a No. 4 general-purpose polished finish.

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3.2.7.1.2 Shower doors. Doors shall be the sliding type. Design of the doors including frame and method of attachment shall be at the manufacturer's option. Panels shall be translucent. Pattern on glass or plastic panels shall be at the manufacturer's option. Finish of door frames and door hardware shall be at the manufacturer's option.

3.2.7.2 Shower valves, heads, and fittings. Showers shall consist of an extended arm with wall escutcheon, water control valves, diverter, and shower head. Shower head shall be in accordance with part C as specified (see 6.1). Bathtub showers shall be one of the type valves specified in part E (see 6.1). Shut-off type anti-scald device shall be provided when specified (see 6.1), and shall conform to performance requirements of ASSE standard 1016.

3.2.7.3 Diverter. The design and location of the diverter, for changing direction of water flow between the shower head and the tube spout, shall be at the manufacturer's option.

3.3 Part B, shower receptors.

3.3.1 Materials, shower receptor. Receptor material shall be porcelain enameled steel, concrete, terrazzo, or plastic as specified (see 6.1). Materials for concrete and terrazzo receptors shall be at the manufacturer's option. Material for plastic receptors shall conform to ANSI Z124.2. Materials for porcelain enameled steel receptors shall conform to ANSI A112.19.4.

3.3.2 Construction.

3.3.2.1 Porcelain enamel receptors. The base metal of receptors shall be one-piece construction or welded construction at the manufacturer's option. Thickness of the receptor base before coating shall be 0.069 inch or greater. The receptor shall be mounted on a steel frame. Total height of the receptor including the support frame shall be 6 inches or greater. The underside of the receptor shall be coated with sound-deadening material 3/16-inch thick or greater (see 4.4.1.1).

3.3.2.2 Concrete and terrazzo receptors. Receptor shall be completely molded with floor sloping toward drain and with threshold curbs at the outer edges. Inside corners and edges shall be rounded. Thickness of receptors adjacent to the drain shall be 1-1/2 to 2 inches. Sides for units used in permanent installations shall be 6 $\pm$1/2 inches in height and 3 $\pm$1/4 inches in height for temporary installations. The type of installation, permanent or temporary shall be specified (see 6.1).

3.3.2.3 Plastic receptors. Plastic receptors shall conform to ANSI Z124.2 (see 4.4.2.2).

3.3.3 Surface finish, shower receptors.

3.3.3.1 Porcelain enamel receptors. Exposed inner surfaces shall be coated with acid resisting porcelain enamel. The floor area shall be slip resistant. Pattern and color of the slip-resistant surface shall be at the manufacturer's option. Threshold shall be porcelain enameled, baked enameled, or chromium plated with a bright or satin finish, at the manufacturer's option.

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3.3.3.2 Concrete and terrazzo receptors. Surfaces of concrete and terrazzo receptors shall be resistant to water, grease and ordinary cleaning compounds. Surfaces in way of bathing area shall be slip resistant.

3.3.3.3 Plastic receptors. Surface finish on plastic receptors shall be in accordance with ANSI Z124.2.

3.3.4 Drain. Copper alloy drains shall have a bright chromium plated finish. Cast iron drains shall have a bright or satin chromium plated finish. Stainless steel drains shall have a No. 4 general purpose polished finish. Plastic drains shall be plated or as molded.

3.3.5 Joining surface. Joints between receptors and cabinet walls shall provide an effective seal against leakage under shower bathing conditions.

3.3.6 Fasteners. When fasteners are used they shall be nonferrous metal or stainless steel that is compatible with the joining surfaces.

3.3.7 Color. Color of porcelain enameled steel receptors, and plastic receptors shall be white, and color of concrete and terrazzo receptors shall be mottled white or gray. Value of white and gray shall be at the manufacturer's option.

3.3.8 Performance requirements of receptors.

3.3.8.1 Porcelain enamel and masonry receptors. When tested in accordance with 4.4.2.1 there shall be no cracking on the floor surface of porcelain enamel receptors and no cracking on concrete and terrazzo floor surfaces.

3.3.8.2 Plastic receptors. Plastic receptors shall meet all performance requirements in ANSI Z124.2 (see 4.4.2.2).

3.4 Part C - shower head.

3.4.1 Material, shower head. Body of shower heads, and the body of a separate flow control fitting when applicable, shall be copper alloy, stainless steel, die cast zinc alloy or all molded plastic. Face plate and the internal parts of shower head and flow control fitting shall be copper alloy or copper-nickel alloy, die cast zinc alloy, stainless steel, molded plastic, or any suitable material. Handles on shower heads shall be stainless steel, copper alloy, die cast zinc alloy or molded plastic.

3.4.2 Type I.

3.4.2.1 Mounting. Shower heads shall be one or two of the following mountings, as specified (see 6.1):

(a) Ball joint.
(b) Without ball-joint.
(c) Ball joint and head integral with a formed wall plate.
(d) Shower head integral with a formed wall plate.
(e) Shower head, adjustable height mounted, rigid tube or pipe on swivel.
(f) Shower head, adjustable height mounted, flexible tube.

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3.4.2.1.1 Shower heads mounted on rigid tube or pipe. Length of tube or pipe extension from swivel shall be as specified (see 6.1). Construction of swivel and fittings for shower head shall be at the manufacturer's option.

3.4.2.1.2 Shower heads mounted on flexible tube. Length of tube shall be as specified (see 6.1). Fittings and shower head shall conform to requirements of ASSE standard No. 1014.

3.4.2.1.3 Dimensional tolerances. Dimensional tolerances for rigid and flexible mounted showers shall be at the manufacturer's option.

3.4.2.2 Class 1, nonadjustable spray shower head. The spray face shall be changeable from the normal showering position to a flushing position that permits removal of sediment.

3.4.2.3 Class 2, adjustable spray shower head. The shower head shall be capable of adjustment from a gentle cone spray in accordance with 3.4.2.10.2 to a straight spray and a flushing position that permits removal of sediment.

3.4.2.4 Water saving shower heads. When specified (see 6.1), class (1) and (2) shower heads shall deliver a maximum flow rate of 3.0 gpm at 80 pounds per square inch (psi). Water saving features shall be designed to prevent falling out or causing interference with thread connection to the supply pipe. The fitting shall be quiet in operation and maintain a constant flow. Water-saving features shall be tamper resistant.

3.4.2.5 Shutoff control fittings. When specified (see 6.1), shower heads shall include a flow control feature to temporarily shut off the water supply at the shower head. When a flow control fitting is required, it shall be integral with the shower head or a separate unit as specified (see 6.1).

3.4.2.6 Inlet and supply. Inlet of shower heads and supply pipe shall have 1/2 inch NPT threads. Length and shape of supply pipe shall be as specified (see 6.1).

3.4.2.7 Diameter. The diameter of circle enclosing the face end of the shower head shall be not less than 1 inch or greater than 3-5/8 inches.

3.4.2.8 Color, plastics. Color of plastic shower heads and exposed plastic parts of shower heads shall be black or white.

3.4.2.9 Finish. Stainless steel heads shall have a No. 4 general purpose polished finish. Chromium plated metal and plastic heads, and supply pipe or tube shall have a bright finish. Exposed uncoated parts of plastic may have a molded finish.

3.4.2.10 Performance.

3.4.2.10.1 Maximum operating temperature. Shower heads shall be capable of operating at a temperature of 130° ±5°F when tested in accordance with 4.4.3.1.

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3.4.2.10.2 Type I - spray pattern and discharge rate. Shower heads shall give an even conical spray pattern having a spread of 2-1/2 ±1 foot in a 6 foot drop at any pressure from 20 to 80 psi when tested in accordance with 4.4.3.2.1.

3.4.3 Type II - heads for emergency showers.

3.4.3.1 Shower head. Emergency shower heads shall have a face plate that can be easily removed for cleaning.

3.4.3.2 Diameter. Diameter of shower head shall not be greater than 11 inches.

3.4.3.3 Inlet and supply. Inlet of shower head shall have either 1 or 1-1/2 inch NPT female threads. Size of supply pipe shall be compatible with shower head supplied. Length and shape of supply pipe shall be as specified (see 6.1).

3.4.3.4 Control valve. The shower head unit shall include a self-closing control valve of the same NPT size as the shower head. A chain or rod with a formed handle shall be provided for opening the control valve.

3.4.3.5 Finish. Emergency shower heads shall have one of the following finishes:

(a) Bright chromium plate.
(b) Satin chromium plate.
(c) Lead coating.
(d) Baked enamel or epoxy finish.
(e) As cast aluminum finish.
(f) As cast copper alloy finish.
(g) Nickel plate on a cast surface.
(h) As molded plastic finish.

Chain or rod shall have a bright chromium finish.

3.4.3.6 Performance. Shower heads that are mounted overhead shall have an even vertical spray pattern having a 3-foot minimum spread in a 7-foot vertical drop. All emergency shower heads shall be capable of passing 35 gpm minimum (see 4.4.3.2.2).

3.5 Part D, emergency eye and emergency eye and face wash.

3.5.1 Material, part D. Emergency eye and emergency eye and face wash bowls shall be stainless steel, enameled cast iron, aluminum alloy, vitreous china, or acryionitrile-butadiene-styrene (ABS). Spray nozzles, and valves shall be stainless steel or copper alloy at the manufacturer's option. Stand shall be zinc-coated steel pipe. Type of steel for pipe shall be at the manufacturer's option. When hose attachment is included, the hose shall be any suitable rubber or rubber-like material.

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3.5.2 General, types I and II.

3.5.2.1 Mounting, stainless steel receptors. Stainless steel receptors shall be unmounted, wall mounted, counter top mounted, or pedestal mounted, as specified (see 6.1). When specified (see 6.1), pedestal mounted receptors shall be freeze resistant.

3.5.2.2 Wall mounted receptors. Brackets for wall mounted receptors shall be cast aluminum alloy, cast copper alloy, or chromium plated or enameled cast iron at the manufacturer's option.

3.5.2.3 Pedestal mounted receptors. Receptors shall be mounted on a 1-1/4 or 1-1/2 inch zinc-coated steel pipe at the manufacturer's option. Weight of zinc coat shall be at the manufacturer's option. When freeze resistance is required the valve shall be opened by a handle or foot control, as specified (see 6.1).

3.5.2.4 Receptor thickness. Thickness of stainless steel receptors shall be 0.030 inch or greater (see 4.4.4.1). Thickness of enameled iron, aluminum alloy, plastic, and vitreous china receptors shall be at the manufacturer's option.

3.5.2.5 Inlet and drain. Inlet shall be 3/8 or 1/2 inch, and drain shall be 1-1/4 or 1-1/2 inch IPS at the manufacturer's option.

3.5.3 Type I, eye wash fixtures.

3.5.3.1 Aluminum alloy-receptors. Aluminum alloy receptors shall be wall mounted, or pedestal mounted, as specified (see 6.1).

3.5.3.2 Vitreous china receptors. Vitreous china receptors shall be wall mounted.

3.5.3.3 Plastic receptors. Plastic receptors shall be unmounted, counter top mounted, wall mounted, or pedestal mounted, as specified (see 6.1).

3.5.3.4 Receptor and component parts. Each fixture shall consist of an eye wash receptor, two spray outlets, and a ball or butterfly valve. The valve shall be operated by foot or hand control, as specified (see 6.1). Hand control shall be push bar or flat plate that can be pushed to pen position with the hand, elbow, or arm. When spray hose is required (see 6.1), the valve on the hose shall open by a squeeze down lever handle.

3.5.3.5 Spray outlet. Spray outlet shall direct flow of water into both eyes simultaneously. The outlets shall deliver a minimum of 2 gpm of water through volume controls, aerators or strainers to reduce flow velocity. Shape of spray outlet shall be at the manufacturer's option.

3.5.4 Type II, eye and face wash fixtures.

3.5.4.1 Enameled cast iron receptors, mounting. Enameled cast iron receptors shall be unmounted, wall mounted, or pedestal mounted, as specified (see 6.1). When specified (see 6.1), pedestal mounted receptors shall be freeze resistant.

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3.5.4.2 Receptor. Each receptor shall be complete with fittings. The receptor shape shall provide a spray pattern 64 square inches or greater.

3.5.4.3 Fittings and valve. Each fixture shall include outlets volume regulator and a ball or butterfly valve. The valve shall be operated by a foot or hand control as specified (see 6.1).

3.5.4.4 Spray outlet. Spray shall be through spherical outlets, in-line outlets, or circular ring outlets, or any other suitable arrangement. Spray shall deliver water through volume controls, aerators or strainers to reduce flow velocity. Flow rate shall be 10 gpm or greater when tested in accordance with 4.4.4.2.

3.6 Part E, valves.

3.6.1 Material, mixing valves, and separate valves. Valve body shall be any suitable copper alloy. Internal parts shall be copper or nickel alloy, stainless steel, plastic, or any other suitable material. Material for handles on all valves shall be at the manufacturer's option.

3.6.1.1 Water control and admittance. Valves shall be capable of adjusting water to any desired shower bath temperature with hot water inlet temperature varying from 120° to 180°F and cold water inlet temperature varying from 48° to 70°F. Type T and P valves shall be equipped with an adjustable maximum limit stop or other means of limiting discharge temperature to a predetermined safe value.

3.6.1.2 Interchangeability. Valves of the same type and size, and of the same manufacturer shall be identical to the extent necessary to assure interchangeability of accessories, components, assemblies, and repair parts.

3.6.1.3 Escutcheons. Size of escutcheons shall be the manufacturer's standard size for the valve provided.

3.6.1.4 Finish, valves, and escutcheons. All exposed parts of valves and escutcheons shall be of the following finishes:

(a) Nonferrous and ferrous metals, chromium plated with a bright finish.
(b) Stainless steels with a No. 4 general-purpose polished finish.
(c) Plastic parts shall have a molded, or a bright chromium plated finish.

3.6.2 Types T, P, and M.

3.6.2.1 Size and capacity. Inlet and outlet sizes, and capacity in gpm shall be in accordance with table I.

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TABLE I. Size and capacity of mixing valves and single control valves.

Type	Size, IPS		Capacity 1/ flow rate gpm	Tolerance capacity	
	Inlet inches	Outlet inches		plus gpm	minus gpm
M	1/2	1/2 or 3/4	6	2	2
P	1/2	1/2	6	2	2
T	1/2	1/2	6	2	2
T	1/2	1/2	10	2	2
T	1/2	3/4	15	2	2
T	3/4	3/4	20	2	2
T	3/4	3/4	25	3	3
T	3/4	1	45	5	5
T	1	1-1/4	60	6	6
T	1-1/4	1-1/4	70	7	7
T	1-1/4	1-1/4	80	8	8
T	1-1/4	1-1/2	85	9	9
T	1-1/4	1-1/2 or 2	125	13	13
T	1-1/2	1-1/2 or 2	130	13	13
T	2	2	150	15	15
T	2	2	200	20	20
T	2	2	225	20	30

1/ Capacities rated at 45 psi inlet pressure or 45 psi differential pressure under flow conditions. For recommended usage range, consult manufacturer's literature or catalogs.

3.6.2.2 Stuffing box. Stuffing box for valves shall be packed by one of the following methods at the manufacturer's option:

(a) Spring loaded permanently packed.
(b) Lantern ring packing.
(c) Preformed packing ring.
(d) O-ring.

3.6.2.3 Hand control. Valves shall have control settings of cold, warm and hot or indicate direction of handle rotation to increase temperature at the manufacturer's option. Markings shall be clear and permanent.

3.6.2.4 Type T, thermostatic valves. Valves shall be furnished with automatic thermostatic controls. The thermostatic control shall be accessible for setting and calibration without disturbing pipe or tube connections.

3.6.2.5 Type P, pressure balancing valves. Valves shall be furnished with a pressure balancing mechanism that responds immediately to fluctuations of hot and cold water pressure. The pressure balancing mechanism shall be accessible without disturbing pipe or tube connections.

3.6.2.6 Type M, mechanical mixing, single control. Single control mixing valves shall have a single control to turn water on and off and regulate temperature of flow. Operating parts subject to wear shall be removable for replacement without disconnecting the valve from the supply lines, or without using special tools. Cartridge type valve shall be constructed so that all of the operating parts subject to wear are within the cartridge and can be replaced as a unit.

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3.6.2.7 Type C, combined supply, separate hot and cold water valves. Valves shall have stems that are replaceable along with threaded barrel or bonnet. Stem packing or seals shall be renewable. Valves shall be compression or replaceable cartridge type. Compression type valves shall have renewable seats.

3.6.3 Performance, valves.

3.6.3.1 Type T and P valves. Type T and P valves shall meet the performance requirements of ASSE standard No. 1016 (see 4.4.5).

3.6.3.2 Type M and C valves. Type M and C valves shall meet the performance requirements of ANSI A112.18.1 (see 4.4.5).

3.7 Part F, drains.

3.7.1 Drains. Drains shall be shower, area, floor or trench as specified (see 6.1).

3.7.2 Outlet. Drains shall have an inside caulk bottom drain outlet, outside caulk side drain outlet, or plain end outlet as specified (see 6.1). Plain end outlet shall be bottom drain, or side drain, as specified (see 6.1).

3.8 Workmanship.

3.8.1 Shower cabinets, wall mounted showers, and column showers. Porcelain enameled and baked enameled steel showers shall be free from cracks, chips and blisters on exposed areas. Stainless steel wall mounted and column mounted showers shall be free from seams, laps and cracks. Plastic showers shall be free from defects listed in ANSI Z124.2.

3.8.2 Shower and bathtub doors. Glass and plastic panels for shower doors shall be free from cracks and shall not contain any edge cracks.

3.8.3 Shower receptors. Porcelain enameled steel receptors shall be free from cracks, chips, and blisters on exposed areas. Surfaces of concrete receptors shall be free from pits, holes, and loose aggregate, and surfaces of terazzo receptors shall be free from loose marble chips. Plastic receptors shall be free from defects listed in ANSI Z124.2.

3.8.4 Shower heads. Stainless steel shower heads shall be free from seams, and cracks. Chromium plated copper alloy shower heads shall be free from cracks. Chromium plated plastic heads and parts, and plastic parts of shower heads shall be free from cracks and mold defects affecting performance and appearance.

3.8.5 Emergency eye, and emergency eye, and face wash. Metal parts shall be free from cracks and sharp edges. Vitreous china and enameled cast iron shall be free from chips and cracks. Plastic receptors shall be free from cracks and mold defects affecting performance and appearance.

3.8.6 Valves. Valve bodies shall be free from cracks, cold shuts and clustered surface porosity. Parting line shall be flush except adjacent to flanges. Sharp edges shall be removed. Castings shall not be plugged or repair welded.

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3.8.7 Drains. Drains shall be free from cracks, cold shuts and clustered surface porosity. Openings in drains shall be unobstructed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection (see WW-P-541/GEN).

4.2 Sampling for examination and tests. In accordance with WW-P-541/GEN, and as specified herein.

4.3 Examination and tests.

4.3.1 Examination. Examine showers, eye and face wash outfits, and other items selected in accordance with 4.2 for the defects listed in table II. The inspection level shall be as specified in WW-P-541/GEN. The AQL shall be 2.5 percent defective for major defects and 4.0 for minor defects.

TABLE II. Classification of defects, end item.

Defects	Major	Minor
Shower installations outfits and shower stalls (cabinets only).		
Type and class not as specified.	X	
Material not as specified	X	
Soap dish not included on cabinet showers.		X
Panel mounted showers not as specified.	X	
Control valves and shower heads for column mounted showers not as specified.	X	
Glass and plastic panels for bathtub and cabinet showers cracked.	X	
Cracks, chips, blisters on exposed areas of porcelain enameled and baked enameled cabinets and showers.	X	
Cracks, seams and laps in wall mounted and column mounted showers.	X	
Finish not as specified.	X	
Defects greater than those listed in ANSI Z124.2 (for plastic showers only).	X	
Shower receptors.		
Material not as specified.	X	
Underside of porcelain enameled units uncoated.		X
Standing area not slip resistant.		X
Defects greater than those listed in ANSI Z124.2 (for plastic receptors only).	X	
Cracks, chips and blisters on exposed area of porcelain enameled receptors.	X	
Pits, holes, and loose aggregate in concrete receptors.	X	
Stone chips loose in terrazzo receptors.	X	
Shower heads.		
Type not as specified.	X	
Mounting not as specified and diameter not as specified.	X	
Flow control, and control fittings not as specified.	X	

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TABLE II. Classification of defects, end item - cont'd.

Defects	Major	Minor
Finish on emergency showers not as specified.		X
Cracks, seams and laps in stainless steel heads, cracks in chromium plated heads, and cracks in molded plastic heads and plastic parts of heads.	X	
Surface of chromium plated plastic heads wavy.		X
Emergency eye, and face wash.		
Cracks in metal parts.	X	
Sharp edges on metal parts.		X
Cracks or chips in vitreous china or enameled cast iron.	X	
Cracks in plastic receptors.	X	
Valves.		
Type not as specified.	X	
Internal parts of valves not as specified.	X	
Diverter not as specified.	X	
Inlets and outlet size not as specified.	X	
Cracks, cold shuts and clustered surface porosity in valve bodies.	X	
Parting line not flush and sharp edges present.		X
Valve body plugged or repair welded.	X	
Drains.		
Cracks and cold shuts in drains.	X	
Clustered surface porosity in drains		X
Drain openings obstructed.	X	

4.3.2 Tests. Samples selected in accordance with 4.2 shall be tested as specified in WW-P-541/GEN and as specified herein. The AQL for tests shall be as specified in WW-P-541/GEN.

4.4 Test methods.

4.4.1 Thickness, shower installations.

4.4.1.1 Thickness, shower cabinets, and receptors. Determine thickness as follows (see 3.2.2.3, 3.2.2.4, 3.2.2.5, and 3.3.2.1):

(a) Porcelain enamel and base metal. Determine thickness of porcelain using a suitable magnetic gage. Determine thickness of base metal by measuring the total thickness using a micrometer or suitable caliper and subtract thickness of porcelain enamel.
(b) Enamel paint and base metal. Measure total thickness using a micrometer or suitable caliper and determine thickness of base metal and enamel paint after measuring the thickness of the enamel using a magnetic or scratch gage.

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(c) Pilasters, posts, rails, columns, access panels and receptor flange. Measure thickness using a micrometer or any other suitable measuring device.
(d) Receptor with sound deadening material on underside. Measure depth using any suitable depth gage or measuring device.

4.4.1.2 Shower curtain rod and shower rail, shower cabinets, and bathtub showers. Measure diameter of tubes using a suitable caliper and measure thickness using a tube micrometer with measuring surfaces suitable for curved surfaces (see 3.2.2.7.1 and 3.2.7.1.1).

4.4.1.3 Panel mounted and column showers. Measure thickness using a suitable micrometer, caliper, ultrasonic pulse, or frequency unit (see 3.2.3.2.1, 3.2.3.2.2, 3.2.4, 3.2.5.2, 3.2.5.3, 3.2.5.4, and 3.2.6.1.3).

4.4.2 Receptors and plastic showers.

4.4.2.1 Porcelain enamel, concrete and terrazzo receptors. Secure a nipple and elbow to the drain and a horizontal pipe approximately 30 inches long. Position the receptor to permit suspension of a 50 pound weight at a distance of 24 inches to the vertical center of the drain. When 50 pound weight is suspended there shall be no cracking (see 3.3.8.1).

4.4.2.2 Plastic showers and receptors. Test in accordance with ANSI Z124.2 (see 3.3.2.3 and 3.3.8.2).

4.4.3 Shower heads.

4.4.3.1 Hot water test. Flow water at 130° +5°F for 1 hour through shower heads and observe whether they perform satisfactorily (see 3.4.2.10.1).

4.4.3.2 Water spread and flow rate.

4.4.3.2.1 Type I:

(a) Procedure. Connect shower head to pipe or tubing of the same nominal size as the shower head inlet. The pipe or tubing shall be at least 20 inches inside diameter long at the fitting inlet. Pressure measurement device(s) shall be located 1/2 to 2-1/2 inches inside diameter upstream from the shower head inlet. Shower head valve(s) shall be at maximum flow setting.
(b) Test. Position shower head above a drain in the center of a gradually sloped surface. Open valve(s) and maintain flow pressure at 20 pounds per square inch gage (psig). Measure and record spray pattern dimensions and discharge rate. Repeat test at 80 psig and record results. Determine conformance to 3.4.2.10.2.

4.4.3.2.2 Type II. Adjust water flow rate to 35 gpm through shower head positioned 7 feet above a drain in the center of a gradually sloped surface. Measure spread of water cone near or at the floor surface and observe whether the cone provides a full even shower (see 3.4.3.6).

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4.4.4 Emergency eye, and emergency eye and face wash.

4.4.4.1 Eye, and eye and face wash fixtures. Measure thickness of stainless steel receptor using a suitable micrometer, caliper, or ultrasonic pulse or frequency unit (see 3.5.2.4).

4.4.4.2 Emergency eye and face wash. Thoroughly flush the supply, then pressurize it between 15 and 18 psi. Determine rate of flow using an inserted flow meter on the supply or a suitable receptacle at the drain and a stop watch. Observe the receptor to determine whether it drains water satisfactorily and does not overflow (see 3.5.4.4).

4.4.5 Valves. Test valves in accordance with ASSE standard No. 1016 or ANSI A112.18.1 as applicable (see 3.6.3.1 and 3.6.3.2).

5. PREPARATION FOR DELIVERY (see WW-P-541/GEN)

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Part, type, and class required (see 1.2.1 and 1.2.2).

Shower installations:

(c) Size of steel and plastic cabinets (see 3.2.2.1).
(d) Curtain rod or door and entry (see 3.2.2.7 and 3.2.2.7.3).
(e) Concealed or exposed cabinet fittings (see 3.2.2.8).
(f) Arrangement of multiple shower partitions, number, and location of shower units (see 3.2.3.5).
(g) Arrangement of combination shower and dressing room partitions, number and location of shower and dressing units, and whether additional supports for seats are to be included (see 3.2.3.6).
(h) Emergency shower construction (see 3.2.4).
(i) Single shower unit design (see 3.2.5.1).
(j) Soap dish or liquid soap dispenser (see 3.2.5.2 and 3.2.5.3).
(k) Number of shower stations for wall mounted inline showers (see 3.2.5.3).
(l) Number of shower stations for back-to-back inline shower panels, mounting and soap dish or liquid soap dispenser (see 3.2.5.4).
(m) Control valves, mounting and type (see 3.2.5.5).
(n) Type of column shower and number of dressing rooms (see 3.2.6).
(o) Column height (see 3.2.6.1.1).
(p) Removable cover plate or extension tube for top of columns and ceiling height (see 3.2.6.1.5).
(q) High capacity dome gate, when required (see 3.2.6.1.6).
(r) Control valves and mounting for column showers (see 3.2.6.2).

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Bathtub showers.

(s) Sliding doors or curtain rails (see 3.2.7.1).
(t) Shower heads and valves, required (see 3.2.7.2).
(u) When anti-scald device required (see 3.2.7.2).

Shower receptors.

(v) Material required (see 3.3.1).
(w) Whether permanent or temporary type required for concrete and terrazzo receptors (see 3.3.2.2).

Shower heads.

(x) Type I shower mounting (see 3.4.2.1).
(y) Length of tube or pipe for mounted adjustable height shower heads (see 3.4.2.1.1 and 3.4.2.1.2).
(z) Water saver shower head, where required (see 3.4.2.4).
(aa) Flow control fittings, when required, and whether integral or separate (see 3.4.2.5).
(bb) Length and shape of supply pipe required (see 3.4.2.6 and 3.4.3.3).

Eye, and eye and face wash.

(cc) Type mounting required (see 3.5.2.1, 3.5.3.1, 3.5.3.3, and 3.5.4.1).
(dd) When freeze resistant pedestal mounted receptors are required (see 3.5.2.1 and 3.5.4.1).
(ee) Type control required for freeze-resistant fountains (see 3.5.2.3).
(ff) Valve operation, and spray hose when required for type I (see 3.5.3.4).
(gg) Type valve operation required for type II (see 3.5.4.3).

Drains.

(hh) Type of drain (see 3.7.1).
(ii) Type outlet and drain location (see 3.7.2).

6.2 Shower cabinet size. Prior to placing orders, purchasers should consult suppliers concerning dimensional requirements.

6.2.1 Shower curtains. Plastic shower curtains are covered by L-C-7&0, and curtain rings or hooks are covered by MIL-S-955.

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Army - ME
Navy - YD
Air Force - 99

Review activities:

DLA - CS

User activities:

Army - CE
Navy - MC, MS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOE - AEC
GSA - FSS
HUD - TCS

Preparing activity:

Navy - YD

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